X Health AI Platform

Conversation Intelligence Layer

Hard Configuration — Product Design Document

Design Philosophy

Most health apps treat users as symptoms to be solved. We treat them as people to be understood. Every data point we collect from a conversation exists for one reason: to make the next interaction more human, more useful, and more trustworthy than the last. This document defines the intelligence architecture that makes that possible — and why each layer exists.

1. Why This Matters — The Intelligence First Principle

When a user opens X and types or says something, they are giving us a gift: a raw, unfiltered signal of what they need right now. Not what we think they need. Not what their demographic suggests. What they actually need, in this moment, in their own words.

The Conversation Intelligence Layer is our system for reading that signal with precision. Every conversation is run through a structured extraction pipeline that classifies, tags, and stores psychological and behavioural data points — not for surveillance, but for service.

These data points are hard configurations. They are designed and maintained by us. Clients cannot alter them. This is intentional: the quality of user understanding must be consistent, medically ethical, and immune to commercial pressure.

The Core Insight

A user who says 'can asthma kill you?' is not always asking a clinical question. They might be newly diagnosed and terrified. They might be a parent watching their child wheeze. They might be a veteran patient who just had a scare. The words are the same. The need is completely different. Our intelligence layer tells us which one we're talking to.

2. The Intelligence Stack — What We Collect

Every conversation processed by X is analysed across eight distinct intelligence dimensions. Each dimension is extracted automatically from the conversation transcript using the LLM and stored as structured data. Together, they form a psychological and behavioural portrait of the user interaction.

Dimension	Classification Type	Primary Purpose
1. Persona Archetype	Enum (8 types)	Understand who the user is behaviorally
2. Primary Intent	Enum (15 types)	Understand what they are asking for
3. Secondary Intent	Enum (7 modifiers)	Understand the context layer beneath
4. Emotional State	Enum (10 states)	Understand how they feel right now
5. Innate Desire	Enum (8 categories)	Understand what they actually need
6. Health Literacy	Enum (3 levels)	Understand how to speak to them
7. Engagement Signal	String (multi-value)	Understand how to bring them back
8. Response Quality	Enum (3 levels)	Understand how well we served them

3. Dimension Deep Dives

3.1 Persona Archetype

What it answers: Who is this user, behaviourally?

Persona is the highest-order signal. It tells us the user's relationship with their condition, their information-seeking style, and their emotional baseline — all inferred from how they write or speak, not from a form they filled out.

Why it matters: The same clinical question needs a completely different response depending on who is asking. 'What triggers asthma?' asked by The Newly Diagnosed needs foundational reassurance and simple language. Asked by The Information Seeker, it needs mechanism depth and references. Asked by The Anxious Worrier, it needs emotional grounding first, information second.

Archetype	Behavioural Signals	What the AI adjusts
The Newly Diagnosed	'just found out', 'don't know much', basic questions, fear/confusion	Foundational language, heavy reassurance, no jargon
The Veteran Patient	Medical terminology, long history references, nuanced questions	Peer-level depth, respect prior knowledge, skip basics
The Anxious Worrier	'what if' questions, catastrophising, repeated fear mentions	Emotional grounding first, shorter turns, calming cadence

Archetype	Behavioural Signals	What the AI adjusts
The Skeptic	Questions medical advice, 'natural' alternatives, pharma distrust	Non-defensive tone, evidence without pressure, respect autonomy
The Caregiver	'my child', 'my mother', third-person questions	Acknowledge caregiver burden, practical focus, empathy for their role
The Information Seeker	Rapid detailed questions, mechanism interest, research references	Substantive depth, structured responses, link to further learning
The Action Taker	'what can I do', 'give me steps', prevention focus	Step-by-step format, practical and immediate, forward-looking
The Emotional Processor	Shares personal story, expresses frustration/grief/burnout	Space before substance, validation-first, no unsolicited advice

Product Implication

Persona is the master signal that governs response style, cadence, and depth across all other dimensions. It is extracted once per session and stored. Over time, repeat users build a persona history that informs longitudinal engagement strategy.

3.2 Primary Intent

What it answers: What is the user asking for?

Intent is not the same as the words the user used. It is the underlying ask beneath those words. A user who says 'I feel short of breath after walking' may be asking SYMPTOM_UNDERSTANDING, or they may be in EMOTIONAL_SUPPORT mode (scared and looking for normalisation), or they may be preparing for a DOCTOR_PREP conversation. Intent classification determines the response architecture.

Intent	What the user wants	AI response mode
LEARN_BASICS	Foundational education	Structured, simple, progressive
UNDERSTAND_TRIGGERS	Trigger-specific knowledge	Specific, actionable, concrete
MEDICATION_EDUCATION	How medications work (not what to take)	Mechanistic explanation, no prescription
SYMPTOM_UNDERSTANDING	What symptoms mean (not diagnosis)	Normalise, contextualise, redirect if needed
LIFESTYLE_MANAGEMENT	Diet, exercise, environment, routines	Practical, personalised, step-oriented

Intent	What the user wants	AI response mode
EMERGENCY_GUIDANCE	Severe/acute situation	Immediate, concise, escalate instantly
CAREGIVER_SUPPORT	Helping someone else with the condition	Empathetic, role-aware, dual focus
EMOTIONAL_SUPPORT	Anxious, scared, overwhelmed	Validation first, information second
DOCTOR_PREP	Preparing for medical appointment	Structured prep, question scaffolding
DEVICE_EDUCATION	Inhalers, spacers, nebulizers etc.	Practical how-to, visual language
MYTH_BUSTING	Common misconception	Gentle correction, non-condescending
SEEK_DIAGNOSIS	Asking for a diagnosis	Hard redirect — must not comply
SEEK_PRESCRIPTION	Asking what medication to take	Hard redirect — must not comply
SMALL_TALK	Off-topic conversation	Warm, brief, redirect gently
UNKNOWN	Intent unclear	Ask one clarifying question

Hard Rule

SEEK_DIAGNOSIS and SEEK_PRESCRIPTION are the only intents that trigger a mandatory hard redirect. The AI must never attempt to fulfil these, regardless of how the question is framed. This is non-negotiable and cannot be overridden by soft config.

3.3 Secondary Intent

What it answers: What is the contextual modifier beneath the primary intent?

Secondary intent is an optional but high-value signal. It adds a layer of context that changes how the primary intent should be served. The same primary intent of LIFESTYLE_MANAGEMENT reads very differently when the secondary intent is CHILD_CONTEXT versus ELDERLY_CONTEXT.

Secondary Intent	What it signals	How it modifies the response
COMPARISON	User is comparing two options	Balanced presentation, no steering
PERSONAL_STORY	User is sharing their experience	Active listening mode, reflect before informing
CHILD_CONTEXT	Question is about a child	Age-appropriate language, parental concern acknowledgment
ELDERLY_CONTEXT	Question is about an older person	Simpler medication refs, comorbidity

Secondary Intent	What it signals	How it modifies the response
		awareness
NEWLY_DIAGNOSED	Recent diagnosis (user or someone close)	Extra patience, no assumption of prior knowledge
LONG_TERM_PATIENT	Years of condition management	Respect expertise, skip fundamentals
SEEKING_REASSURANCE	User wants to feel calmer	Validation before information, always

3.4 Emotional State

What it answers: How does this user feel right now?

Emotional state is the most time-sensitive signal. Unlike persona (which is relatively stable) or intent (which is session-specific), emotional state can shift within a single conversation. It governs the AI's tone, pace, sentence length, and whether it leads with information or with empathy.

We extract emotional state from: word choice, punctuation patterns, question phrasing, self-referential language, and intensity markers.

Emotional State	What it looks like in text	AI cadence response
CALM_CURIOUS	Relaxed phrasing, exploratory questions, no urgency	Balanced, exploratory, can include depth
MILDLY_ANXIOUS	Some worry words, seeking normalisation	Reassure early, then inform
HIGHLY_ANXIOUS	Fear language, catastrophising, fast questions	Ground first. Very short turns. No information overload
FRUSTRATED	'nothing works', 'I'm tired of this'	Validate without minimising. No unsolicited advice
OVERWHELMED	'too much', 'I can't cope', topic overload	One thing at a time. Reduce cognitive load
HOPEFUL	Positive framing, seeking progress	Match energy, affirm, reinforce positive trajectory
DEFEATED	'given up', 'nothing helps', hopelessness	Gentlest tone. Do not push information. Human warmth first
URGENT	Fast typing signals, acute situation described	Immediate, no preamble, safety first
NEUTRAL	No strong emotional signal	Standard, balanced response

Emotional State	What it looks like in text	AI cadence response
GRIEVING	Processing diagnosis, loss, worsening condition	Hold space. Do not rush to solutions

Critical Rule

HIGHLY_ANXIOUS, DEFEATED, and URGENT states trigger automatic cadence shifts in the AI. The Wit Layer and Philosophy Layer are disabled. Sentence length is capped. Emotional validation is mandatory before any clinical content. These rules cannot be modified by client soft config.

3.5 Innate Desire (The Beneath-the-Question Layer)

What it answers: What does this user actually need, beneath what they asked?

This is the most psychologically sophisticated dimension in the stack — and the most commercially powerful. Innate desire is not the same as intent. Intent is what the user is asking for. Innate desire is why they're asking, and what would make them feel genuinely served.

A user asking 'what are the long-term effects of steroid inhalers?' might have the intent MEDICATION_EDUCATION. But their innate desire might be FEAR_RELIEF (they've been prescribed them and are scared). Or CONTROL (they want to understand what they're putting in their body). Or VALIDATION (their doctor told them it's safe and they want to believe it).

Knowing the desire transforms the response from correct to resonant.

Innate Desire	What the user really needs	How it changes the response
FEAR_RELIEF	To be told this is not as bad as they think	Normalise. De-catastrophise. Clinical facts framed reassuringly
CONTROL	To understand so they can act	Mechanistic depth. Actionable options. Empower the decision
VALIDATION	To feel their concern is legitimate	Acknowledge first. 'You're right to ask this.' Never dismiss
CONNECTION	To not feel alone with this condition	Normalise prevalence. 'Many people experience this'
SIMPLICITY	To have the complexity reduced for them	One clear answer. No branching. No 'it depends'
PROGRESS	To know things are improving or can improve	Hopeful framing. Trajectory language. Small wins
SAFETY	To know what the danger signs are	Clear, calm safety guidance. No alarmism, no minimisation

Innate Desire	What the user really needs	How it changes the response
IDENTITY	To understand how this condition fits their life	Personal framing. 'People with your profile often find...'

Why This Is the Most Valuable Signal

Intent data tells you what people want to know. Innate desire data tells you what your product actually needs to do for them to feel helped. This is the layer that separates an answer engine from a health companion. It is the foundation of longitudinal engagement — because users come back to things that make them feel understood, not just informed.

3.6 Health Literacy Level

What it answers: How should we speak to this user?

Health literacy is inferred from vocabulary precision, question specificity, and prior knowledge demonstrated in the conversation. It is not a permanent label — it is recalibrated per session and can shift mid-conversation.

Literacy Level	How we identify it	Language calibration
Beginner	Lay vocabulary, broad questions, 'what is' framing, confusion signals	Analogies, zero jargon, short sentences, one idea at a time
Intermediate	Knows condition name, tracks some metrics, uneven knowledge	Mix of plain language with occasional precise terms, explained in context
Advanced	Medical terminology used correctly, asks mechanism-level questions, references own history	Peer-level language, respect prior knowledge, no condescension

Literacy Drift

A user may present as Intermediate in session one and Advanced in session three. We track literacy across sessions to personalise over time. A user who is growing in health literacy is a user who is engaging — and that engagement signal is a product health metric.

4. Additional Intelligence Dimensions — Beyond the Baseline

The four dimensions above (persona, emotion, innate desire, literacy) form the core stack. But thinking from first principles — about what would make a health AI product genuinely extraordinary — there are four more dimensions we should collect. Each one unlocks a capability that no health AI currently offers at scale.

4.1 Conversation Arc (Emotional Trajectory)

Most analytics capture a snapshot — how the user felt at the start. Arc captures movement — how the user's emotional state evolved across the session. Did they start anxious and end reassured? Did they start hopeful and become confused? Did the AI fail to shift their state at all?

Arc Pattern	What it signals	Product action
Anxious → Reassured	AI successfully de-escalated	Flag as model session for training
Neutral → Hopeful	Engagement layer working	Track affirmation effectiveness
Anxious → Still Anxious	AI failed to de-escalate	Flag for review; improve MILDLY_ANXIOUS handling
Hopeful → Confused	Information overload likely	Reduce response density for this intent
Frustrated → Validated	Emotional processing worked	Replicate tone pattern in similar sessions

Why This Matters

Arc is the single best proxy for session quality. A successful clinical answer that leaves the user more anxious than before is a failure. A session that shifts the user from anxious to grounded — even without perfect clinical information — is a success. Arc is how we measure the emotional work the AI does.

4.2 Readiness to Act

A user who is emotionally ready to change behaviour has very different needs from one who is still processing. Readiness to Act is adapted from behavioural change theory (Stages of Change / Transtheoretical Model) and identifies where the user sits on the action spectrum.

Stage	Behavioural signals	What X should do
Pre-contemplation	Not considering change; may be in denial	Do not push. Build awareness gently
Contemplation	Aware, weighing options, 'I should probably'	Provide comparative information, remove barriers to action
Preparation	Ready to act, asking 'how do I start'	Give concrete first steps, specific and small

Stage	Behavioural signals	What X should do
Action	Actively changing, tracking, engaging	Reinforce with affirmation, track progress signals
Maintenance	Sustaining change, asking about relapse	Help normalise setbacks, reinforce identity as 'someone who manages this'

Commercial Implication

Readiness to Act is the most direct predictor of a user's appetite for features like reminders, tracking, and care team integration. A user in the Preparation stage is the highest-value acquisition moment. This dimension tells you when to prompt and when to wait.

4.3 Trust Signal

Trust is the invisible currency of health AI. Users who trust the platform engage more, share more, and are more likely to follow through. Trust signal measures the user's expressed confidence in the AI across a session.

Extracted from: direct statements ('is this accurate?', 'can I trust this?'), hedging behaviour, counter-questions, requests for sources, and acceptance of AI guidance without pushback.

Trust Level	Behavioural indicators	AI response adaptation
High Trust	Accepts information without challenge, asks follow-up, thanks the AI	Continue current tone and depth
Neutral Trust	No explicit trust signals either way	Maintain; include one soft credibility anchor per session
Low Trust	Questions accuracy, mentions 'but I read that', asks for sources	Acknowledge uncertainty honestly, never over-claim, cite reasoning
Active Distrust	Challenges AI identity, expresses frustration with AI advice	Name the limitation openly. Suggest human support. Do not escalate claims

Why This Is Strategic

Trust erosion is the leading cause of health app abandonment. Users do not stop using health apps because the information is wrong. They stop because they feel the app does not understand them or is overpromising. Trust Signal measurement is early warning infrastructure for churn.

4.4 Engagement Depth Score

Engagement depth is a composite signal that measures the quality of a user's involvement in a session — not just how long they stayed, but how deeply they engaged.

Inputs: number of follow-up questions after first answer, depth of question evolution (moving from broad to specific), personal disclosure level, self-referential language frequency, return visit rate, and session-end sentiment.

Depth Level	What it indicates	Platform action
Surface	Single question, accepted answer, no follow-up	Good for info retrieval. Low stickiness signal
Moderate	2-4 follow-up questions, some personal disclosure	Engaged user. Candidate for memory feature introduction
Deep	Personal story, multi-turn exploration, questions evolve	High-value user. Return trigger should be activated
Vulnerable	Emotional disclosure, health anxiety, personal fear sharing	High-care user. Gentlest engagement layer. Potential escalation watch

5. What the Full Intelligence Picture Looks Like

When all eight dimensions are captured in a single session, we have a portrait of the user that no other health AI platform currently builds. Here is what that portrait looks like in practice.

Example: Session Portrait

Persona: The Anxious Worrier | Primary Intent: EMOTIONAL_SUPPORT | Secondary Intent: SEEKING_REASSURANCE | Emotional State: MILDLY_ANXIOUS → Reassured | Innate Desire: FEAR_RELIEF | Health Literacy: Intermediate | Readiness to Act: Contemplation | Trust Signal: Neutral → High | Engagement Depth: Moderate What this tells us: A user who came in scared, needed to feel heard before they needed to feel informed, was ready to consider change but not act yet, and left trusting the platform more than when they arrived. Next session: memory callback, gentle readiness-to-act nudge, no affirmation overload.

This portrait is not demographic data. It is not a user profile filled out in onboarding. It is real-time behavioural intelligence derived entirely from how this specific person spoke to us in this specific session. It is earned understanding — and it compounds across every conversation.

6. What We Do With It — Aggregate Intelligence

Individual session portraits drive per-user personalisation. Aggregated across thousands of sessions, they become a product intelligence engine that no clinical survey could replicate.

What aggregate intelligence unlocks:

- Which intents correlate most strongly with session abandonment (product gap identification)
- Which emotional states lead to the highest trust gain (AI quality benchmarking)
- Which persona archetypes have the highest return visit rate (engagement ROI)
- Which innate desires are most underserved by current responses (content gap analysis)
- Which health literacy level has the worst resolution rate (language calibration signal)
- When in a condition journey users are most emotionally receptive (intervention timing intelligence)
- Which emotional arcs predict churn (early warning system)

The Strategic Asset

This aggregated dataset — structured, longitudinal, and psychologically rich — is a category-defining asset. It is the kind of data that health systems, researchers, and payers have never had access to. The question of how we steward it, share it, and protect it is as important as the question of how we collect it.

7. Ethical Architecture — How We Collect Without Exploiting

Every intelligence dimension in this document is extracted for one purpose: to make X more useful to the user we're talking to. The moment any of this data is used to manipulate, upsell, or exploit vulnerability, we have broken the foundational promise of the product.

The following rules are non-negotiable and embedded in hard config:

1. Emotional distress signals (HIGHLY_ANXIOUS, DEFEATED, GRIEVING) are never used to drive commercial actions. They trigger care mode, not marketing mode.
2. Innate desire data is never used to create false urgency or exploit fear. FEAR_RELIEF desire means we reduce fear — not amplify it to drive engagement.
3. Persona and desire data is never surfaced to clients in a form that could identify individual users.
4. The AI must never misrepresent itself as human or as a clinician, regardless of trust level or engagement depth.
5. Users must always be able to access human support. The AI is a companion, not a gatekeeper.
6. Intelligence collection is disclosed in plain language to users. We collect this to serve them better. That is the only reason.

The Test

For every intelligence dimension we collect, ask: 'If a user found out we were tracking this, would they feel helped or surveilled?' If the answer is anything other than 'helped', the dimension does not belong in this system.

8. Summary — The Complete Intelligence Stack

Dimension	Type	Config	Core Value Unlocked
Persona Archetype	Enum (8)	Hard — Admin only	Response style personalisation
Primary Intent	Enum (15)	Hard — Admin only	Response architecture selection
Secondary Intent	Enum (7)	Hard — Admin only	Contextual response modulation
Emotional State	Enum (10)	Hard — Admin only	Tone and cadence adaptation
Innate Desire	Enum (8)	Hard — Admin only	Resonance over correctness
Health Literacy	Enum (3)	Hard — Admin only	Language calibration
Conversation Arc	Composite	Hard — Admin only	Emotional quality measurement
Readiness to Act	Enum (5)	Hard — Admin only	Behavioural intervention timing
Trust Signal	Enum (4)	Hard — Admin only	Churn prediction and prevention
Engagement Depth	Enum (4)	Hard — Admin only	Return-visit and stickiness signals

Closing Note

The ambition of X is not to be the most accurate health information service. Accuracy is table stakes. The ambition is to be the health companion that makes people feel genuinely understood — session after session, condition after condition, year after year.

This intelligence stack is how we build that. Not through surveillance. Through attention.

Every dimension we collect is an act of listening. Listening carefully enough to respond well. And responding well enough that people come back.